

Video Analysis Report

Why Fighter Jets Ban 90% of C++ Features

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Video Analysis Report

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Core Analysis

Claims Analysis

ARGUMENT SUMMARY:

Safety-critical aerospace software like F-35 uses restricted C++ (JSF standard) that removes dangerous features like exceptions, dynamic memory allocation, and recursion to ensure reliability.

TRUTH CLAIMS:

CLAIM 1: Ariane 5 rocket crashed on June 4, 1996 due to unhandled software exception.

SUPPORT: The European Space Agency’s official inquiry confirms the Ariane 5 Flight 501 failure on June 4, 1996, was caused by an overflow error when converting a 64-bit floating point number to a 16-bit signed integer in the Inertial Reference System software. The failure occurred 36.7 seconds after launch, destroying the rocket and its payload valued at approximately \$500 million. IEEE Spectrum and other technical publications documented this as one of the most expensive software bugs in history.

REFUTATION: The specific timing of “36 seconds” is slightly imprecise; official reports state 36.7 seconds. Cost figures vary depending on whether development costs are included, though the order of magnitude is generally accepted.

RATING: A (Definitely True)

LABELS: Factual, technical, historical, well-documented

CLAIM 2: F-35 JSF C++ standard prohibits use of exceptions (AV rule 208).

SUPPORT: The publicly available “Joint Strike Fighter Air Vehicle C++ Coding Standards” document (Lockheed Martin Corporation, December 2005) explicitly states in AV Rule 208: “C++ exceptions shall not be used.” This document is widely cited in aerospace software engineering literature and is publicly accessible online. Multiple aerospace software engineering textbooks reference this specific prohibition.

RATING: A (Definitely True)

LABELS: Factual, technical, verifiable, documented

CLAIM 3: The first microprocessor was created by Garrett AiResearch for F-14, not Intel.

SUPPORT: The Garrett AiResearch CADC (Central Air Data Computer) used in the F-14 was developed in the late 1960s and incorporated microprocessor technology. Ray Holt documented the MP944 chip set was operational before Intel’s 4004. The technology was classified until 1998.

REFUTATION: The definition of “microprocessor” is contested. Intel’s 4004 (1971) is widely recognized as the first single-chip microprocessor, while the Garrett system used a multi-chip architecture. The Computer History Museum and most authoritative computing histories credit Intel with the first commercial microprocessor. The claim depends on how “microprocessor” is defined—single-chip versus multi-chip implementation, commercial versus military application.

LOGICAL FALLACIES: Equivocation—using different definitions of “microprocessor” to make the claim.

RATING: C (Medium)

LABELS: Contested, definitional, technical, requires-context

CLAIM 4: F-4 Phantom had essentially no software, using mechanical bombing computers instead.

SUPPORT: Historical documentation confirms early F-4 variants used analog mechanical computing systems for weapons delivery. The AN/ASQ-91 weapons release computer system used in early F-4s was largely mechanical with cams and gears. DoD historical documents describe early 1960s-1970s fighter aircraft as having minimal digital computing capability.

REFUTATION: Later F-4 variants (particularly F-4E and beyond) incorporated digital systems and avionics computers. The claim oversimplifies by not distinguishing between early and late F-4 variants produced over a 20+ year period.

LOGICAL FALLACIES: Overgeneralization—treating all F-4 variants as identical when discussing software content.

RATING: B (High)

LABELS: Generally-accurate, requires-nuance, historical, technical

CLAIM 5: F-35 contains 9 million lines of onboard code.

SUPPORT: Multiple defense industry sources and Lockheed Martin documentation cite figures of 8-9 million lines of code for the F-35’s onboard systems. A 2009 IEEE Spectrum article cited approximately 5.7 million lines for the aircraft

and 3.5 million for ground systems, totaling over 9 million. Various defense publications and congressional testimonies reference similar figures.

REFUTATION: The exact number varies by source and may depend on what software is counted (onboard only vs. total system including ground support). Some sources cite different figures ranging from 8-24 million lines when including all related systems. Software lines of code continue to grow with updates, making any specific number time-dependent.

RATING: B (High)

LABELS: Generally-accurate, approximate, technical, contextual

CLAIM 6: Department of Defense mandated Ada usage for approximately 15 years starting in 1980s.

SUPPORT: DoD Directive 3405.1 (1987) and DoD-STD-2167 required Ada for mission-critical systems unless a waiver was granted. The Ada mandate was officially in place from the mid-1980s until it was lifted in 1997 with the revision to MIL-STD-498. Numerous government and academic sources document this approximately 12-15 year period of Ada mandate.

REFUTATION: The mandate was lifted in 1997, making the duration closer to 10-12 years depending on when enforcement truly began, though “approximately 15 years” is reasonable. The mandate had numerous waivers and exceptions even during its enforcement period.

RATING: A (Definitely True)

LABELS: Historical, factual, documented, government-policy

CLAIM 7: Bjarne Stroustrup contributed to formulation of JSF C++ rules.

SUPPORT: Stroustrup has publicly acknowledged involvement in discussions about the JSF C++ coding standards in interviews and writings. The JSF coding standards document acknowledgments and various aerospace software engineering publications reference consultation with C++ language experts including Stroustrup.

REFUTATION: The exact extent of Stroustrup’s involvement is not fully documented in publicly available sources.

RATING: B (High)

LABELS: Likely-accurate, expert-involvement, qualified-claim

CLAIM 8: Ada compilers cost thousands of dollars in the 1990s.

SUPPORT: Historical documentation shows commercial Ada compilers in the 1980s-1990s were expensive, often costing several thousand dollars per license. This was common for specialized language compilers before the open-source movement gained momentum.

REFUTATION: GNAT (GNU Ada Translator) became available as a free Ada compiler in the mid-1990s, though commercial versions still cost money. The timeframe when “all” Ada compilers were expensive was relatively brief before free alternatives emerged.

LOGICAL FALLACIES: Hasty generalization—suggesting all Ada compilers were expensive without noting free alternatives that emerged.

RATING: B (High)

LABELS: Generally-accurate, historical, economic, contextual

OVERALL SCORE:

- LOWEST CLAIM SCORE: C
- HIGHEST CLAIM SCORE: A
- AVERAGE CLAIM SCORE: B+

OVERALL ANALYSIS:

The argument presents well-researched technical and historical claims about aerospace software engineering with strong supporting evidence. Most claims are factually accurate and verifiable. Weaknesses include occasional oversimplification and definitional ambiguities. Strengths include specific references to documented standards and historical events.

Technology Impact Analysis

SUMMARY

JSF C++ coding standard demonstration showing how removing language features creates safer aerospace software through strict rules eliminating exceptions, dynamic memory allocation, and recursion for F-35 fighter jet programming.

TECHNOLOGIES USED

- C++ programming language with JSF (Joint Strike Fighter) coding standard
- X-Plane 12 flight simulator
- Web API for real-time flight data streaming
- Python for GUI frontend development
- Custom Multi-Function Display (MFD) hardware
- Real-time data processing and calculation systems

- Embedded systems programming for aerospace applications

TARGET AUDIENCE

- Aerospace software engineers and defense contractors
- Military aviation programs (US Air Force, Navy, international partners)
- Safety-critical systems developers
- Flight simulator enthusiasts and researchers
- Computer science students learning embedded systems
- Aviation industry professionals requiring certified software

OUTCOMES

- Demonstrates practical implementation of JSF C++ standard for safety-critical aerospace applications through live flight simulation integration.
- Provides open-source educational repository showing compliant versus non-compliant code examples for learning purposes.
- Illustrates historical transition from Ada to C++ in military aviation software development and Pentagon approval process.
- Shows real-time data processing capabilities connecting flight simulator to custom hardware display using restricted C++ subset.
- Educates developers on memory safety, exception handling elimination, and recursion restrictions in mission-critical software environments.

SOCIETAL IMPACT

Positive Impacts: The project democratizes understanding of safety-critical software development by making JSF coding standards publicly accessible and demonstrable. It contributes to aerospace safety by promoting coding practices that prevent catastrophic failures like the Ariane 5 disaster. Educational value helps train future aerospace engineers in proper safety-critical development methodologies. Open-source approach fosters transparency in military software development practices, enabling broader community contribution and review.

Negative Impacts: Focuses exclusively on military applications, potentially normalizing weapons systems development. May inadvertently provide technical insights useful for adversarial purposes despite using unclassified information. Creates perception that restrictive coding practices are only necessary in aerospace, potentially limiting adoption in other safety-critical civilian applications. Resource-intensive development approach may not be accessible to smaller organizations or developing nations.

ETHICAL CONSIDERATIONS

The project presents medium ethical concerns. Primary issues include military application focus (F-35 fighter jet) designed for combat operations, raising questions about contributing knowledge to weapons systems. However, the educational intent and use of publicly available standards mitigate severity. The project promotes safety and transparency in software development, which are positive ethical contributions. No direct harm is demonstrated, and the

simulator-based approach avoids actual weapons development. The open-source educational framework democratizes knowledge rather than restricting it to defense contractors.

SUSTAINABILITY

Environmental Perspective: Limited direct environmental impact as software-focused educational project. However, promotes efficiency in aerospace systems which could reduce fuel consumption through optimized flight control. Military aviation applications have inherent environmental costs through fuel consumption and emissions. Software optimization may contribute marginally to reducing computational resource requirements.

Economic Perspective: Highly sustainable economically through open-source model requiring minimal ongoing investment. Educational value creates skilled workforce for aerospace industry. Demonstrates cost-effective alternatives to proprietary Ada compilers that historically cost thousands. Knowledge sharing reduces duplicated development efforts across defense contractors. However, remains dependent on expensive flight simulator software and specialized hardware.

Social Perspective: Strong social sustainability through educational mission and knowledge democratization. Reduces barriers to aerospace software development understanding. Creates community of practice around safety-critical systems. Long-term value in training developers who can work across safety-critical industries including medical devices and autonomous vehicles. Potential concern regarding military focus limiting broader civilian applications adoption.

SUMMARY AND RATING

Educational aerospace software project demonstrating safety-critical C++ coding practices through flight simulation, promoting transparency and knowledge sharing while focusing on military applications with moderate ethical considerations.

- Societal Benefit: MEDIUM
- Sustainability: MEDIUM

Key Extractions

Wisdom

SUMMARY

Lori demonstrates programming F-35 fighter jets using restricted C++ following JSF standards, emphasizing safety-critical aerospace software development principles.

IDEAS

- Fighter jet software failures can destroy half a billion dollars in an instant catastrophically.
- F-35 programming prohibits exceptions entirely to prevent catastrophic failures during critical flight operations.
- Modern military aircraft are essentially flying computers that happen to also be planes.
- The first microprocessor was secretly created for F-14 Tomcat, predating Intel's public release.
- F-35 contains 9 million lines of code, showing exponential growth in aerospace software complexity.
- Pentagon mandated Ada language use for 15 years to standardize military software development.
- Ada compilers cost thousands of dollars, making C++ and GCC attractive for student learning.
- Lockheed Martin convinced Pentagon to break their own law by using C++ instead of Ada.
- JSF standard surgically removes dangerous C++ features while maintaining language power and flexibility.
- Joint Strike Fighter standard is publicly available online for anyone to read and study.
- Remove-before-flight concept means eliminating language features rather than adding safety mechanisms.
- Programming for safety-critical systems requires quantifying what you remove, not what you add.
- Memory allocation, exceptions, and recursion are three main safety concerns in aerospace C++ programming.
- F-4 Phantom's bombing computer was literally just gears and cams, containing no software.
- F-14's swing wing required automated control because manual pilot control was too demanding.
- Military used over 450 different programming languages simultaneously before Ada standardization effort.
- X-Plane flight simulator connects via web API for real-time aircraft data integration testing.
- Safety-critical code must be 100% reliable 100% of the time without exceptions.
- Bjarne Stroustrup himself helped formulate the JSF C++ rules for fighter jet programming.
- Predictable behavior matters more than powerful features in mission-critical aerospace software development.
- Multiple countries and contractors collaborated on F-35, requiring public coding standards for cooperation.
- Flight control software processes polynomial equations through physically burned microcode in early systems.
- Ariane 5 rocket exploded because unhandled exception converted 64-bit float to 16-bit integer.

- Early military branches used completely incompatible languages: Jovial for Air Force, CMS-2 for Navy.
- Student developers chose free C++ over expensive Ada, creating workforce incompatibility with military.

INSIGHTS

- Safety-critical programming succeeds by restricting language features rather than adding protective layers.
- Catastrophic software failures stem from allowing operations the system cannot handle rather than bugs.
- Modern warfare depends more on reliable software execution than physical aircraft capabilities.
- Standardization emerges from practical workforce realities, not just technical superiority alone.
- Exponential software complexity growth forces fundamental rethinking of programming approaches and language design.
- Public accessibility of military coding standards enables broader collaboration while maintaining security.
- Predictability and reliability matter infinitely more than flexibility when failure means death.
- Language restrictions create safety by making impossible what should never happen rather than catching errors.
- Historical technology secrecy often hides true innovation timelines, reshaping our understanding of progress.
- Economic factors like compiler costs shape entire generations of programmers and industry standards.
- Military software evolution mirrors broader computing trends but with decades-long commitment cycles.
- Real-time systems demand zero-tolerance approaches where consumer software accepts occasional failures.
- Convincing institutions to break their own rules requires demonstrating superior alternatives with concrete benefits.
- Safety culture emerges from removing temptation rather than trusting discipline to avoid dangerous practices.
- Collaborative international projects require transparent standards despite traditional military secrecy.

QUOTES

- “At Mach 1, you don’t have time for a garbage collector in most programs.”
- “When failure isn’t an option, what happens when one bad line can be catastrophe?”
- “It’s not about what’s in the code, it’s about quantifying what you take out.”
- “Half a billion dollars destroyed in an instant by a single unhandled exception thrown.”

- “Exceptions shall not be used in F-35 programming under any circumstances whatsoever.”
- “Modern military aircraft are more about software than hardware, fundamentally speaking about capabilities.”
- “The first microprocessor was created for F-14 years before Intel, but kept secret.”
- “F-35 has 9 million lines of code onboard, an exponential trajectory of complexity.”
- “Department of Defense used over 450 different programming languages with no proper standards.”
- “Ada compilers cost thousands of dollars, making free